

Problem 12.526 — Jordan Canonical Form & GM

Aryansingh Sonaye
AI25BTECH11032
EE1030 - Matrix Theory

November 3, 2025

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Let A be a 7×7 matrix such that

$$2A^2 - A^4 = I_7 \quad (1.1)$$

where I_7 is the identity matrix. If A has two distinct eigenvalues and each eigenvalue has geometric multiplicity 3, then the total number of nonzero entries in the Jordan canonical form of A equals:

- (A) 6 (B) 7 (C) 8 (D) 9

Given Data

Symbol	Meaning	Value
A	Square matrix	7×7
Equation	Polynomial relation	$2A^2 - A^4 =$
Eigenvalues	Distinct eigenvalues of A	Two distinct (λ
Geometric multiplicity (GM)	$\dim(\ker(A - \lambda I))$	3 for each eigen
Total GM	Sum over both eigenvalues	$3 + 3 = 6$

1. Geometric Multiplicity

For a square matrix A and eigenvalue λ ,

$$\text{GM}(\lambda) = \dim(\ker(A - \lambda I)). \quad (3.1)$$

It represents the number of linearly independent eigenvectors for λ .

If the **algebraic multiplicity (AM)** equals GM, then A is diagonalizable; otherwise A is **defective** and has Jordan blocks larger than 1×1 .

2. Jordan Canonical Form

Every square matrix A is similar to a matrix J such that

$$A = PJP^{-1} \quad (3.2)$$

where J is the **Jordan canonical form** and P has columns that are eigenvectors and generalized eigenvectors.

A Jordan block of size k for eigenvalue λ is:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}. \quad (3.3)$$

The 1 above the diagonal links a generalized eigenvector to a true eigenvector, e.g.

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad (3.4)$$

with $(A - \lambda I)\mathbf{v}_1 = 0$ and $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$.

3. Determining Block Sizes via Null Spaces

Define

$$N_k(\lambda) = \ker((A - \lambda I)^k). \quad (3.5)$$

These satisfy

$$N_1(\lambda) \subseteq N_2(\lambda) \subseteq N_3(\lambda) \subseteq \cdots \quad (3.6)$$

and their dimensions reveal Jordan structure:

Quantity	Meaning
$\dim N_1(\lambda)$	Number of Jordan blocks (= GM)
$\dim N_2 - \dim N_1$	Number of blocks of size ≥ 2
$\dim N_3 - \dim N_2$	Number of blocks of size ≥ 3

Thus, computing $\ker((A - \lambda I)^k)$ for successive k tells how many blocks of each size exist.

Step 1: Polynomial & Step 2: Eigenvalues

$$2A^2 - A^4 = I_7 \Rightarrow A^4 - 2A^2 + I = 0 \quad (4.1)$$

$$\Rightarrow (A^2 - I)^2 = 0 \quad (4.2)$$

Let λ be an eigenvalue of A .

$$(\lambda^2 - 1)^2 = 0 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1. \quad (4.3)$$

So A has two distinct eigenvalues: $+1$ and -1 .

Step 3: GM Data $\Rightarrow$ Jordan Structure

$$\text{GM}(+1) = 3, \quad \text{GM}(-1) = 3. \quad (4.4)$$

Total eigenvectors = 6. Since A is 7×7 , one dimension is missing — corresponding to one generalized eigenvector.

$\Rightarrow$ There is **one** Jordan block of size 2 and **five** blocks of size 1.

Step 4: Jordan Form

(Assume the 2×2 block belongs to $\lambda = +1$; the count is unaffected.)

$$J = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}. \quad (4.5)$$

Step 5: Count Nonzero Entries

Each 1×1 block contributes one nonzero. The 2×2 block contributes three (two diagonal entries and one superdiagonal):

$$\text{Total nonzeros} = 5 \times 1 + 3 = 8. \quad (4.6)$$

Final Answer

8

Conclusion

The null spaces $\ker((A - \lambda I)^k)$ reveal the Jordan block sizes: each jump in $\dim N_k$ counts blocks of size $\geq k$. Here, only one eigenvector is missing from full diagonalizability $\Rightarrow$ exactly one 2×2 block.

$$\boxed{\text{Total nonzero entries in the Jordan form of } A = 8.} \quad (5.1)$$